

Inequalities – 3

Integral inequalities: the working techniques

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Theoretical minimum.

Theorem 1 (Chebyshev’s integral inequality). If f and g are both non-decreasing, or both non-increasing, on $[a, b]$, then $\frac{1}{b-a} \int_a^b fg \geq \frac{1}{b-a} \int_a^b f \cdot \frac{1}{b-a} \int_a^b g$; if they are oppositely ordered, the inequality reverses. The proof is the sign of $\iint (f(x) - f(y))(g(x) - g(y)) dx dy$.

Theorem 2 (Jensen for integrals). For convex φ and a probability density w on $[a, b]$, $\varphi(\int wf) \leq \int w\varphi(f)$. In particular $(\int_0^1 f)^2 \leq \int_0^1 f^2$, and $\int_0^1 e^f \geq e^{\int_0^1 f}$.

Fact 1 (the four moves). Almost every integral inequality of a student olympiad is one of four moves. Normalise, so that the constraint becomes $\int f = 1$ or $\|f\| = 1$. Write the difference of the two sides as $\int (\text{something})^2$ or as a double integral of a product of two differences with a fixed sign. Integrate by parts to move a derivative onto a factor you control. Or introduce a parameter, bound the expression by something depending on it, and optimise in that parameter at the end. In Differentiable functions – 9 the same problems were attacked through the derivative; here the structure of the integral does the work.

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1. Prove Theorem 1 and Theorem 2, and prove the normalisation principle: an inequality homogeneous of degree zero in f may be proved under any one convenient normalisation.
 2. (SEEMOUS 2011) For an integer $n \geq 1$ let $f: [0, 1] \rightarrow \mathbb{R}$ be non-decreasing. Prove that $\int_0^1 f(x) dx \leq (n+1) \int_0^1 x^n f(x) dx$, and find all non-decreasing continuous f for which equality holds.

3. (SEEMOUS 2016) Let f be continuous and decreasing on $[0, \pi/2]$. Prove that $\int_{\pi/2-1}^{\pi/2} f \leq \int_0^{\pi/2} f(x) \cos x \, dx \leq \int_0^1 f$, and determine when the equalities hold.
4. (IMC 1998) Let $f: [0, 1] \rightarrow \mathbb{R}$ be continuous and such that $xf(y) + yf(x) \leq 1$ for all $x, y \in [0, 1]$. Show that $\int_0^1 f \leq \frac{\pi}{4}$, and exhibit a function attaining equality.
5. (Putnam 1973) On $[0, 1]$ let f have a continuous derivative with $0 < f'(x) \leq 1$ and $f(0) = 0$. Prove that $(\int_0^1 f)^2 \geq \int_0^1 f^3$.